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MDM4U Culminating Game Project "Balance at the Third" Report

Table of contents

- Introduction - Pg. 1
- Game Round Reward Table - Pg. 3
- Theoretical Probability - Pg. 3
- Tree Diagram - Pg. 7
- Theoretical Expected Value - Pg. 7
- Experimental Probability and Experimental Expected Value - Pg. 8
- Analysis of Discrepancies - Pg. 10
- Data Visualization - Pg. 11
- Game Strategy and Early Stopping - Pg. 12
- Game Design - Pg. 13
- Conclusion - Pg. 14

Introduction

This is a probability based card game. A game that progressively increases risks and rewards, its a game mode with low win rate and high returns. In this game, the house will secretly draw one card from a standard 52-card deck. This card remains the same throughout all rounds. Players need to guess which card is this step by step.

The house draws one card at the beginning. The same card is used for all three rounds.

Players place an initial bet before Round 1.

Each round requires a guess:

- Round 1: Guess the suit (Hearts, Diamonds, Spades, Clubs)
- Round 2: Guess whether the card is a number (2–10) or a letter (J, Q, K, A)
- Round 3: Guess the exact card

If a player guesses incorrectly at any stage, they lose all their bets.

If a player guesses correctly, they may choose to:

- Continue to the next round, or
- Fold and keep their current winnings

The Materials Needed are: a standard deck (no jokers), things (for example the chips) to be the point, things for record.

Game Procedure:

1. Setup:
    a. The house chooses one card (won't show to the player and won't change the card).
    b. The players place the initial bets (There will be no more betting steps after).
2. Round 1: Player guesses the suit.
    a. Correct → The player's bets will be multiplied by 2.4. The player can keep their reward and leave the game. Or they can keep playing and join the next round.
    b. Incorrect → The player loses all their points. Game End.
3. Round 2: The player guesses the type of the car (number or letter).

a. Correct → if the player chooses the number card, their bets will be multiplied by 1.3. else if the player chooses the letter card, their bets will be multiplied by 2.925. The player can keep their reward and leave. Or keep playing to the next round.

b. Incorrect → The player loses all their points. Game End.

4. Round 3: Player guesses the specific card.

   a. Correct → The player wins the maximum prize. if the player chooses the number card in round 2, their bets will be multiplied by 16.164. else if the player chooses the letter card in round 2, their bets will be multiplied by 7.184.

   b. Incorrect → The player loses all their points. Game End.

5. Game ends.

Game Round Reward Table

Round	Guess	Win Rate	Multiplier	Reward (10 point as the initial capital)	Stage Expectations
1	Suits	1/4	2.4×	24	0.6
2	Number Card	9/13	1.3×	31.2	0.9
3	Card	1/9	16.164×	504.31	1.796
Round	**Guess**	**Win Rate**	**Multiplier**	**Reward 10 (initial capital)**	**Stage Expectations**
1	Suits	1/4	2.4×	24	0.6
2	Letter Card	4/13	2.925×	70.2	0.9
3	Card	1/4	7.184×	504.31	1.796

Theoretical Probability

Round 1: Guess the Suit

- 4 suits
- Probability of guessing correctly: 1/4
- Probability of guessing incorrectly: 3/4

Round 2: Guess the Number/Letter

- 13 cards in total
- Number cards: 2–10 (9 cards)
- Letter cards: JQKA (4 cards)
- Probability of guessing the number correctly = Probability of guessing the letter incorrectly = 9/13
- Probability of guessing the letter correctly = Probability of guessing the number incorrectly = 4/13

Round 3: Guess the Specific Card

- Suit and type are known.
- Number cards: 9 cards, probability of guessing correctly is 1/9, probability of guessing incorrectly is 8/9
- Letter cards: 4 cards, probability of guessing correctly is 1/4, probability of guessing incorrectly is 3/4

Win Rate Chart			
InGame	Explanation	Percentage probability	Overall Win Rate
Round 1 :	All of 4 Suit have 13	25%	25%

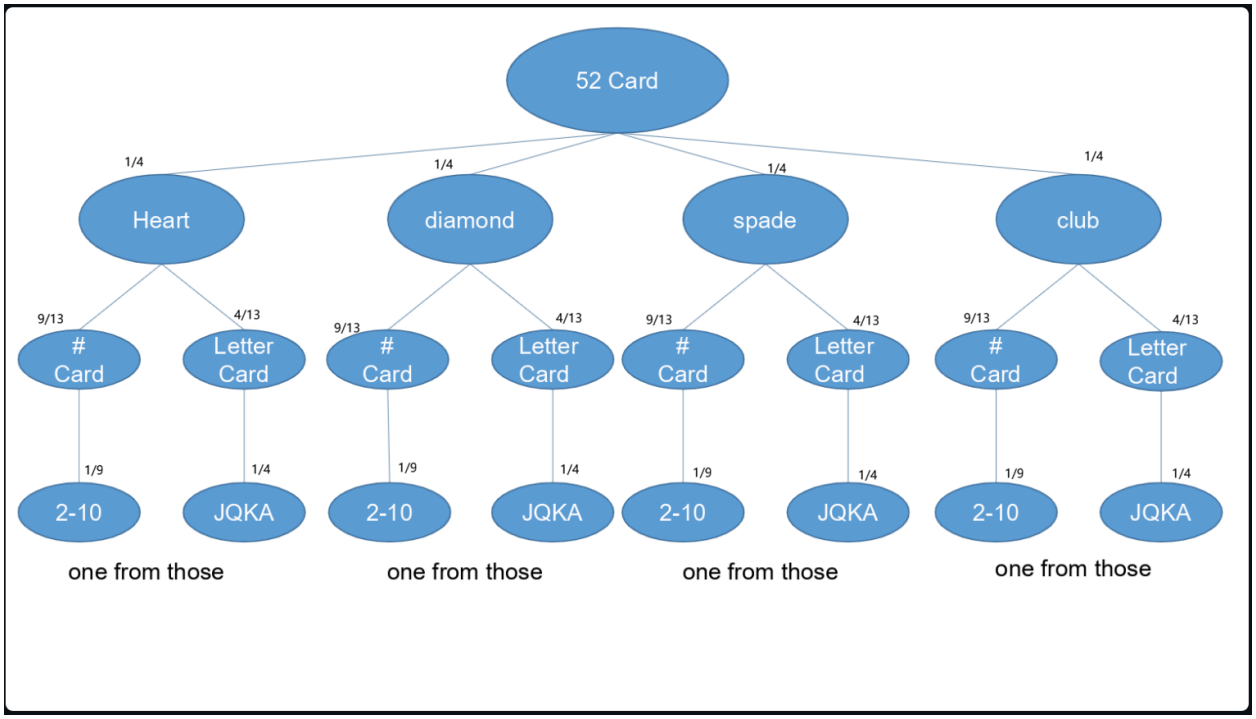
	card, so the probability to guess the suit right is 13/52 = 1/4		
Round 2:	In the 13 cards left, 4 of them are letter cards, they are AKQJ. 9 of them are number cards.	We assume that the player also choose it randomly, the probability is ½ * 9/13 + ½ * 4/13 = 50%	0.25 * 0.5 = 12.5%
Chose Letter in R2:	You have a 4/13 chance to win it.	30.77%	0.25*0.3077 = 7.69%
Chose Number in R2:	You have a 9/13 chance to win it.	69.23%	0.25*0.6923 = 17.3%
Round 3:	the first situation is you chose letter in R2 and win, the second is chose Number in R2 and win	Since the win rate is not the same in Round 2, the probability should be 4/13 * 1/4 + 9/13 * 1/9 = 15.38%	0.125 * 0.1538 = 1.92%

Chose Letter in R2:	you have 1/4 chance to win it.	25%	0.0769 * 0.25 = 1.92%
Chose Number in R2:	you have 1/9 chance to win it.	11.11%	0.173 * 0.1111 = 1.92%

Loss Rate Chart			
InGame	**Explanation**	**Percentage probability**	**Overall Loss Rate**
Round 1:	P(A‘) = 1 - P(A). The P(A) is given in the Win Rate Chart.	75%	75%
Round 2:			Win in R1 and Loss in R2 = 25% * 50% = 12.5%
Chose Letter in R2:		69.23%	25% * 69.23% = 17.31%
Chose Number in R2:		30.77%	25% * 30.77% = 7.69%
Round 3:			Win in R1 and R2, Loss in R3 = 25% * (4/13 * ¾ + 9/13 * 8/9) = 21.15%
Chose Letter in R2:		75%	25% * 30.77% * ¾ = 5.77%

Chose Number in R2:		88.88%	25% * 69.23% * 8/9 = 15.38%

Tree Diagram



Theoretical Expected Value

Based on the above summary of the win rate theory, a 1.92% win rate is far too low. To make this game more like a fair game while still adhering to commercial logic, I designed the game to have a 3% chance of favoring the house. The core idea of this game is to make the EV (Effective Value) negative in the first two rounds, forcing players to enter a third round where they can use a high multiplier to offset the initial losses. This corresponds to the game's name, "Balance at the Third."

EV = Σ(Xn * Pn), Xn is the outcome, Pn is the probability of this outcome to happen. For example look at the Round 1 EV calculation (explanation), $X_1 = 10 \times 2.4 - 10\ coins\ you\ get$, $P_1 = 1\ suit \div 4\ suits = 25\%$, $X_2 =- 10\ coins\ you\ get\ (mean\ you\ loss\ 10\ coins)$,

$P_2 = 3\ suit \div 4\ suits = 75\%$

Then $EV = (10 \times 2.4 - 10) \times 0.25 + (-10) \times 0.75 = -4$.

Full game EV calculation is shown below:

Expected Value (if the initial coin in each round is 10 coin)			
InGame	Explanation	EV	EV return ratio
Round 1:	25% chance get win * (10*2.4 - 10) coins you get + 75% change to get loose * -10 you loss	-4 coin	(10-4) / 10 = 0.6
Round 2:			
Chose Letter in R2:	4/13 chance get win * (10*2.925 - 10) coins you get + 9/13 change to get loose * -10 you loss	-1 coin	(10-1) / 10 = 0.9
Chose Number in R2:	9/13 chance get win * (10*1.3 - 10) coins you get + 4/13 change to get loose * -10 you loss	-1 coin	(10-1) / 10 = 0.9
Round 3:			
Chose Letter in R2:	1/4 chance get win * (10*7.184 - 10) coins you get + 3/4 change to get loose * -10 you loss	7.96 coin	(10+7.96) / 10 = 1.796
Chose Number in R2:	1/9 chance get win * (10*16.164 - 10) coins you get + 8/9 change to get loose * -10 you loss	7.96 coin	(10+7.96) / 10 = 1.796
Overall Expected Value		**10*0.97 = 9.7 coin**	**0.6 * 0.9 * 1.796 = 0.97**

Experimental Probability and Experimental Expected Value

For the experimental probability, I need to get the experiment data first. Therefore I used the programming language Java to write the game, and I gathered nine hundred thousand pieces of data for this. The data I gathered include the game you win, the game you lose in round 1/2/3, the total point you bet, and the total point you won. A part of the code is shown below, and the full version can be found in the github repository.

```java
public static void main(String[] args){
    int success = 0;
    float moneyInput = 0;
    float moneyAward = 0;
    for(int i = 0; i < 900000; i++){
        float v = GameStart();
        moneyInput += 10;
        if(!(v == 0)){
            success += 1;
            moneyAward += v;
        }
    }
    System.out.println();
    System.out.println();
    System.out.println("------Experiment Result------");
    System.out.println("You won " + success + " games in 900000 games");
    System.out.println("There are " + LossInRound1 + " loses in round 1.");
    System.out.println("There are " + LossInRound2 + " loses in round 2.");
    System.out.println("There are " + LossInRound3 + " loses in round 3.");
    System.out.println("You have spend $" + moneyInput + " in total.");
    System.out.println("Your reward is $" + moneyAward + " in total.");

}
```

Below is the data collect,

```
------Experiment Result------
You won 17181 games in 900000 games
There are 674857 loses in round 1.
There are 112799 loses in round 2.
There are 95163 loses in round 3.
You have spend $9000000.0 in total.
Your reward is $8665851.0 in total.

进程已结束，退出代码为 0
```

The experimental win rate (WR) is the game won divided by the total game round number. So WR = 17181 ÷ 900000 ≈ 1.9%. Experimental Expected Rate of Return is (Total revenue - input cost) / Input cost = (8,665,851 coin - 9,000,000 coin) / 9,000,000 coin ≈ -0.0371 = -3.71%. The data is very close to 3% advantage to the house, which means it's accurate. This will also be demonstrated in the data visualization section later (loss rate split bar graph).

Analysis of Discrepancies

Given the massive dataset of 900,000 samples, the theoretical win rate of 1.92% and the experimental win rate of 1.909% are very close, with negligible error. However, the Experimental Expected Rate of Return (-3.71%, representing a 3.71% house advantage) still differs from the theoretically set 3% house advantage by 0.71. Given the inherently high-risk, high-reward nature of the game, coupled with the still-present sampling error, this error is

reasonable. If the number of trials increases indefinitely, the experimental probability will tend towards the theoretical probability.

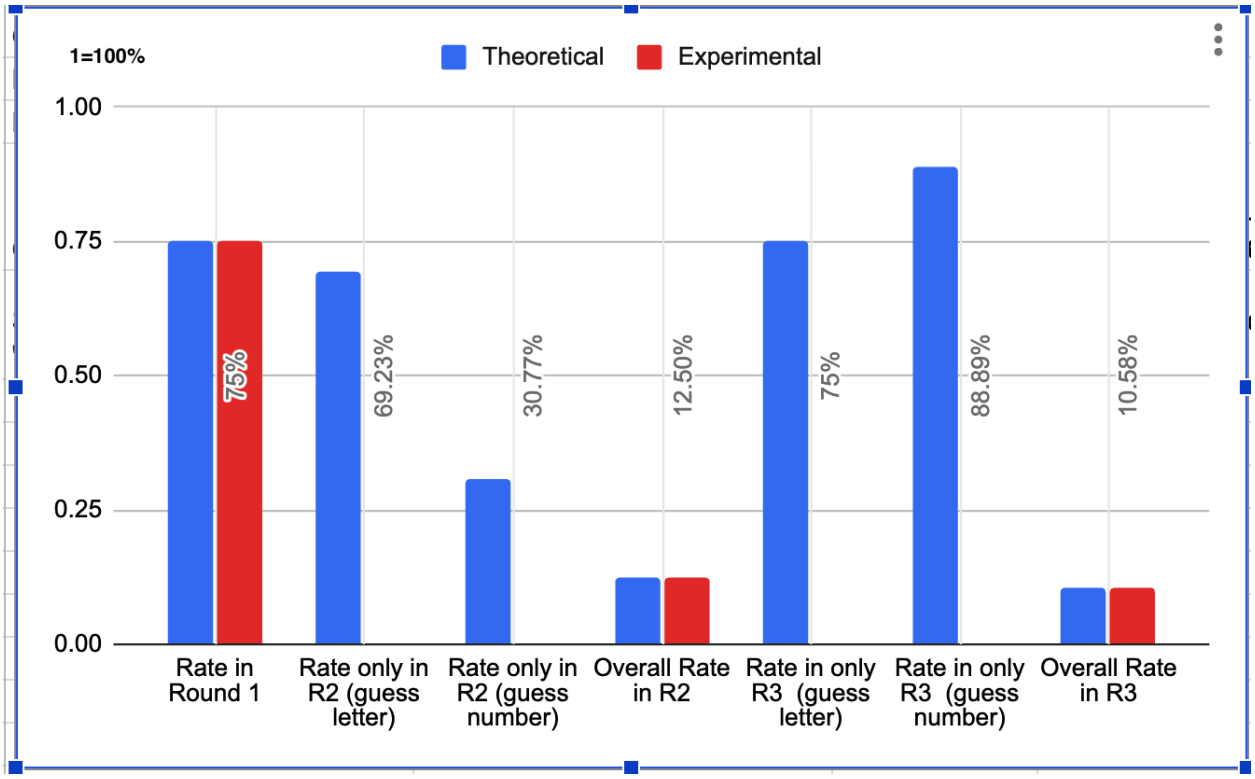
Data Visualization

- Box Whisker Plot

Chart For Box Whisker Plot		
	Theoretical EV	Exprimental EV (using the datas)
Min Value	-10 coin	-10 coin
Q1	-10 coin	-10 coin
Q2	-10 coin	-10 coin
Q3	-10 coin	-10 coin
Max Value	-10 coin	-10 coin
Outlier	494.31 coin	494.31 coin
Explaination: Since the solution that maximizes the expected value is continue play the game to the end. There are only 2 outcomes that will appear. One is you loss all (10 coins) your coins, or you win 503.31 coins. And the win rate is 1.92%. It mean theoretically, we consider 10,000 player, 192 player will win, all of the others will loss. The Box Whisker Plot will be so extreme that the graph have no box on it, because Q1 and Q3 overlap, what should have been a rectangle becomes a horizontal line. IQR = Q3-Q1 = 0， The “max” point on the Box Whisker plot is also -10 coins, 494.31 coin become an outlier.		

- Split Bar Graph (Loss Rate)

Split Bar Graph (Loss Rate)				
	Theoretical	Theoretical	Experimental	Experimental
Rate in Round 1	3/4 = 75%	75%	674857/900000 = 74.98%	74.98%
Rate only in R2 (guess letter)	9/13 = 69.23%	69.23%		
Rate only in R2 (guess number)	4/13 = 30.77%	30.77%		
Overall Rate in R2	1/4(9/13 * 1/2 + 4/13 * 1/2) = 12.5%	12.50%	(100% - 74.98%) * 112799/(900000-674857) = 12.54%	12.54%
Rate in only R3 (guess letter)	3/4 = 75%	75%		
Rate in only R3 (guess number)	8/9 = 88.89%	88.89%		
Overall Rate in R3	1/4 * 4/13 * 1/2 * 3/4 + 1/4 * 9/13 * 1/2 * 8/9 = 10.58%	10.58%	(100% - 74.98%) * (900000-674857-112799)/(900000-674857) * (95163/(900000-674857-112799)) = 10.58%	10.58%
Since we don't know what the players chose from 900,000 data, we have to assume they randomly(50/50) chose between "numbers" and "letters" to get a theoretical number.				



Game Strategy and Early Stopping

In this game, the player is allowed to leave and keep their winning streak after Round 1 or Round 2. They do not have to continue to the next round. At first glance, this seems to give the player a "safer" option and might look like it reduces the house edge.

However, from the expected value perspective, both of the first two rounds have negative EV in terms of net gain:

- Round 1 EV = −4 coins (EV return ratio = 0.6)
- Round 2 EV = −1 coin (EV return ratio = 0.9)

This means that even if the player chooses to stop early after Round 1 or Round 2, the average outcome of those rounds is still a loss compared to the initial bet.

Therefore, Any strategy that involves stopping early does not change the fact that the player is playing a game with negative expected value. The house still makes money in the long run. So that we only need to consider the optimal solution, which is to keep playing until the very end. And we can see there is no real strategy, only probability.

Game Design

This game mode is meticulously designed. Although the theoretical returns reflect the standard house advantage, its core architecture uses 'dynamic expected value gradient' and 'risk aversion trap' to influence players' decisions, ultimately ensuring a higher system profit margin through early exit behavior.

1. Loss Aversion

   Players who won the first two rounds, having experienced the luck and gains of those rounds, tend to shift their focus from maximizing profits (i.e., the expected value of the third round being 1.796) to seeking certainty, due to the general aversion to loss. Players fear losing their existing points in the third round with a low probability of 1/4 or 1/9. This "secure your gains" mentality leads to a large number of players leaving the game after the second round.

2. Utilizing Loss Aversion to Boost the House's Profits

   The overall EV (Effective Value) is 97% of the initial investment, calculated based on the assumption that players will advance to the next round if they qualify. If a player chooses

to leave after the second round, their cumulative EV at that point is initial investment × 0.6 × 0.9. This means that for every additional player who leaves in the second round, the house earns an extra 1 - 0.6 × 0.9 = 1 - 0.54 = 46% of their initial investment as profit. Therefore, if a sufficient proportion of players withdraw due to fear of the third round's risk, the system's actual profit will far exceed 3%! This aligns perfectly with business logic.

Conclusion

Overall, “Balance at the Third” managed to successfully apply knowledge from probability, simulation techniques, and behavioral economics to develop a mathematical framework for creating a commercially attractive game model. The theoretical calculation and experimental simulation tests support the validity of the developed probabilistic model. The game functions properly: it features a very low winning probability, a very high instability, and an unchanging house edge. Moreover, its mechanism uses the predictable nature of certain behavioral biases (e.g., loss aversion and risk sensitivity) to induce second-best decision-making on the part of the player. Which makes the game more in line with commercial logic.